**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Virtual Memory

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

1. What happens to RAM and ROM contents when power is removed?

2. Q = (A AND B) OR C. A=1, B=0, C=1. What is Q?
   A) 1
   B) 0
   C) 2
   D) 10

3. "A Boolean variable represents any unrestricted paragraph of text." Correct this. Explain why.

4. A program joins strings and takes part of an eight-character word. Correct both claims. Explain why. (a) Joining strings adds their numbers. (b) substring(1,3) always means positions 1 to 3 in every language.

## Today's objectives

Explain when a computer needs virtual memory. Describe how the operating system moves pages between RAM and secondary storage.

## Key vocabulary - Read, cover, recall

**Virtual memory**
Secondary storage used as backing space for RAM.

**Page**
Block of memory moved between physical and secondary storage.

**Paging**
Transferring memory pages between RAM and backing storage.

**Thrashing**
Severe slowdown caused by excessive page transfers.

**BLACK PEN: Write one definition in your own words.**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New Learning - Key ideas

**When RAM is insufficient**
More working data may be required than physical RAM can hold at once.

**Virtual memory**
The operating system uses a reserved area of secondary storage as extra backing space for memory pages.

**Moving pages**
Pages are moved between RAM and secondary storage as needed; this is substantially slower than accessing RAM.

**Performance consequence**
Frequent movement between RAM and storage can cause a noticeable slowdown (thrashing).

## I Do - Worked model

**A laptop needs 6 GB for active programs but has only 4 GB available RAM. Explain what virtual memory does and why performance may fall. [4 marks]**

- Identify demand and capacity
- 6 GB is needed but only 4 GB of physical RAM is available.
- The OS can use some secondary storage as backing for inactive pages.
- Accessing secondary storage is slower; swapping pages frequently slows the system. [4]

## We Do - Guided practice

1. A laptop opens many browser tabs. Explain why virtual memory might be used. [3]

2. Explain why adding RAM could reduce use of virtual memory. [3]

3. A student says virtual memory makes a laptop faster than RAM. Evaluate the statement. [4]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Finish the output message when physical RAM is insufficient.**

```
neededGB = 8
freeRAMGB = 6
if neededGB > freeRAMGB then
...
endif
```

## You Do - Independent exam questions

Q2. State what type of storage is used to provide virtual memory. [1]

Q3. Explain two reasons why extensive paging can reduce performance. [4]

Q4. A program needs 5 GB but only 3 GB of free RAM is available. Explain the purpose of virtual memory in this scenario. [3]

## Exit Ticket - True or False?

An algorithm only gives reliable results when its inputs are handled as specified.

**TRUE [ ]** **FALSE [ ]**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

## Do Now - Model answers

1. RAM loses its contents; ROM keeps its contents because it is non-volatile.

2. A - 1

3. True or false.

4. (a) Joining 'AB' and '12' gives 'AB12', with no added space. (b) For 'COMPUTER', substring(1,3) in OCR ERL gives 'OMP': indexes start at 0 and 3 is the length. Follow each language's rules.

## I Do - Explain the model

Identify demand and capacity

6 GB is needed but only 4 GB of physical RAM is available.

The OS can use some secondary storage as backing for inactive pages.

Accessing secondary storage is slower; swapping pages frequently slows the system. [4]

## We Do - Model answers

1. The total working memory demand may exceed RAM [1]; the OS temporarily stores less-used pages on secondary storage [1] and brings them back when needed [1].

2. More active data fits in physical RAM [1], reducing page transfers to secondary storage [1] and therefore decreasing slow accesses [1].

3. Virtual memory extends apparent working capacity [1] and can allow more programs to remain open [1], but its secondary-storage backing is slower than RAM [1], so heavy paging reduces performance [1].

## Code First - Answer

```
neededGB = 8
freeRAMGB = 6
if neededGB > freeRAMGB then
print("May need virtual memory")
endif
```

Expected output: May need virtual memory

## You Do - Self-mark

Q2: Secondary storage, e.g. SSD/HDD [1].

Q3: Pages transferred between RAM and storage [1], requiring additional input/output operations [1]; secondary storage is slower than RAM [1], so applications wait for required data [1].

Q4: Demand exceeds RAM by 2 GB [1]; some pages can be held in secondary storage [1] and restored to RAM when needed [1].

**GREEN PEN: Choose the hardest answer. Correct it on lined paper.**